

Regression -Retesting-Smoke

Regression Testing

Regression Testing is the process of re-verifying a system after changes to ensure that existing functionalities remain unaffected by new code, bug fixes, or updates.

When to Perform

1. During every new software release.
2. After fixing defects or adding new features.
3. Following code refactoring or configuration changes.

Techniques

1. **Retest All:** Executing the entire test suite.
2. **Selective Testing:** Running tests only on modified modules.
3. **Prioritization:** Focusing on critical or frequently used features first.
4. **Impact Analysis:** Identifying and testing all modules potentially affected by the change.
5. **Integration Testing:** Verifying communication between updated modules.

Retesting

Retesting is the specific process of re-executing test cases that previously failed to verify if the identified bugs have been successfully fixed.

Retesting vs. Regression Testing

- **Retesting:** Focuses only on the **failed cases** to confirm the fix.
- **Regression Testing:** Focuses on **related modules** to ensure the fix didn't break existing features.

Smoke Testing

Smoke Testing is the preliminary execution of critical tests on a new software build to verify its basic stability and determine if it is ready for more intensive testing.

Key Objectives

- **Initial Verification:** Performed at the very beginning of the testing cycle.
- **Critical Scope:** Focuses exclusively on the core functionalities of the software package.
- **Build Stability:** Conducted while the build is still unstable to ensure it is functional enough to proceed to the next QA stage.

